

Designing Opportunity Ecosystems: Professional Student Organizations for First-Generation
Students

A Thesis Prospectus in STS 4500

Presented to

The Faculty of the School of Engineering and Applied Sciences

University of Virginia

In Partial Fulfillment of the Requirements for the Degree

Bachelor of Science in Computer Science

By

Tristan Grubbs

November 15, 2025

I. Introduction:

On today's college campus, students are pressured to obtain "employable" skills and create networks before graduation. The topic has been gaining popularity with major institutions like The Wall Street Journal publishing an article regarding the early student recruitment in Ivy League Schools. Specifically, the article cites finance and consulting clubs of these prestigious schools playing a role in this trend. However, this trend is not confined to Ivy League schools. Through higher education, professional-oriented clubs are a pathway to successful careers for students. These shifting priorities have created a new collegiate ecosystem that does not offer equal opportunities to students.

The definition of a first-generation student is highly debated. Ives & Castillo-Montayo said the following:

"Definitions range from being narrow, "students whose parents do not have any postsecondary experience" (Redford & Hoyer, 2017, p. 2), to being broader, "students whose parents have not received a bachelor's degree" (U.S. Department of Education, 2016, p. 1). In 1998, policymakers amended the Higher Education Act (1965/1998) definition to include students from single-parent homes; however, most traditional definitions do not take into account various family structures. It is also typically not clear whether "parent should apply only to biological parents or be broadened to include stepparents, foster parents, and adoptive parents" (Toutkoushian et al., 2018, pp. 2–3, emphasis in original). Toutkoushian et al. (2018) found in a sample of 7,300 students that the number considered first-generation ranged from 22% to 77% based on the researcher's definition." (p. 141)

Although the common thread among definitions is that the student's guardians lack higher education to some significant degree, often ranging from a complete lack to anything below a bachelor's degree. Due to the spectrum of experiences even among first-generation students, the level to which any individual experiences the structural barriers against which they work varies; however, they still see similar barriers among the group. Limited college knowledge, feelings of non-belonging, weaker social capital, and a cultural mismatch are all key structural barriers that first-generation students are likely to see.

The STS research topic I am exploring is "What barriers limit first-generation students' engagement with professional or career-oriented student organizations, and how can institutions of higher education address them?" The technical project I am building is attacking the problem of access to student groups at higher education institutions. My technical project is an example of a way that institutions of higher education can create an ecosystem of equality between first-generation and continuing-education students.

II. Technical Thesis Topic

There are many structural barriers that cause the average University of Virginia student to struggle to discover, join, and participate in a CIO community. These barriers include finding CIOs after the initial activities fair, not having reliable access to information regarding the many CIOs at UVA, and announcements, sign-ups, and general information scattered across GroupMe, Instagram, and emails. This environment creates a clear problem needing a solution; Without a central CIO hub, students will continue to miss opportunities to join CIOs that match their

interests and continue their collegiate success. After researching the topic, my team found that 40% of students found the process of finding a CIO inefficient and confusing. Furthermore, when talking to CIO leader's half said club dates are hard to share and keep updated, which causes significant confusion for both leaders and prospective members. Additionally, two of my four team members are CIO presidents and express these same problems, reinforcing this as a systematic problem in the CIO process.

This is worth studying because the lack of a unified CIO system leaves students disconnected from social and professional communities, which negatively impacts their sense of belonging and professional development. This especially impacts groups that are already threatened by these areas, chiefly, in my research, being first-generation students. For these students, CIO participation acts as a key path to bridge the equity gap between them and their peers, so barriers in access naturally negatively affect them to a higher degree. This makes this not just a user experience problem but a problem of equity and professional development.

As we researched a solution, we focused on the questions "How do CIOs currently recruit prospective members, and what barriers limit their effectiveness?" and "How do students find new CIOs currently, and what barriers do they feel stop them from being more involved?" (Grubbs et al. p. 4). To address these questions, we conducted interviews and surveyed UVA students, particularly making sure we collected information on CIO leaders to properly understand the problem. Once we create an initial solution to the problem, we will go through a second round of research to assess how well we are addressing the environment we are trying to affect. This second round of research will include focused interviews with a flow script to create a consistent process. Members of my team will have a CIO leader and CIO prospective member script and will take notes, some defined and some impromptu, and log success or failure of tasks.

After our interviews, we are combining our data with a quantitative analysis of the success percentage of tasks and a qualitative analysis of user comments, confusion, or suggestions.

By lowering barriers to entry, we are opening opportunities for students to get involved in all CIOs, but some of the most important ways to create equitable environments are through professional CIOs. In Jackson et al. 2024, the writers discuss how students engage in activities that build employability, professional clubs being an example of such. However, the writers also labeled a lack of awareness of what was available as a main blocker of participation. These activities help students build attributes related to success in the workplace, which are not just limited to job-related skills but also include a complex set of skills related to networks and confidence. Simpfinderfer et al. 2025 further supports the thesis that participation in professional activities supports career success. This article linked these activities to a better starting job, which was further linked to a better current job. A better job in this article was determined by factors including pay, happiness, level of fulfillment, and other smaller factors. The article also made it clear that these findings were discovered throughout the full spectrum of student backgrounds. These articles both support the thesis that my technical project will work to support students' success even after their collegiate experience.

III. STS Thesis Topic

My STS topic focuses on the barriers that limit first-generation students' engagement with professional student organizations and how institutions of higher education can build environments that address them. Specifically, the barriers in equity imbalance of access,

awareness, and belonging between first-generation and continuing-education students. This topic has an important sociotechnical nature because much of CIO communication and social outreach is done through technical applications. Therefore, the structure and use of these applications have real-world implications on equity issues related to the topic.

According to Sudbrock et al. 2024, first-generation students face a steeper transition to the new environment of college. The article goes further to conclude that those first-generation students represented a space of necessary targeted support due to the imbalance in social capital between the first-generation students and their peers. Despite these struggles, they bring a wholly different perspective to their continuing-education peers. Abeyta et al. 2025 found that first-generation students were succeeding for community and family as opposed to personal career fulfillment. In Inkelas et al. 2007, we can see how these motivations can be associated with success in college. This study looked at how Living-Learning programs can act as a tight-knit community that smooths the social transition into college for first-generation students. This goes to address the social credit inequality early and was found to have a positive effect on the first-generation students in the program. These articles bring into focus two areas that should be the focus to help mend the inequity facing first-generation students' social credit and ecosystem.

First, using Bronfenbrenner's bioecological systems theory in Demetriou et al. 2017 to analyze the college work of first-generation students. In the microsystem, first-generation students interact with other students, their classroom, and organizations they are a part of. In the mesosystem, students interact with each other and other organizations, and its organizations also interact with each other. Looking further into the exosystem, the policies of the higher education institutions that students are a part of are key. Using this theory, we see the full ecosystem that surrounds students, some parts they interact with directly, and others indirectly. Most importantly

for institutional recommendations is the exosystem, where they create the ecosystem in which students exist. Based on the theory, institutions should build an opportunity ecosystem where students can increasingly build upon themselves and challenge themselves to force growth. This is further supported by Demetriou et al. 2017 which says students must act with their environment in an increasingly complex way to grow. Based on these ideas, creating a true opportunity ecosystem for first-generation students through professional programs means designing structures that actively connect students with networks and career-building spaces that would be hard to reach otherwise. A well-designed opportunity ecosystem ensures professional organizations become an accessible part of first-generation students' environments so they can gain confidence and add to the complex set of skills required to work.

Secondly, using Putnam's lens of bonding and bridging social capital, we see how student organizations produce social capital for first-generation students. Bonding is when students build social capital through strong interpersonal relationships with each other, and bridging building social capital through many looser relationships, building a network of peers. By viewing social capital through this lens, we can see how student organizations help create belonging for first-generation students. Soria et al. 2023 agrees with this idea, saying that first-generation college students come to college with less social capital, so the capital they gain in college creates belonging in their environment and has measurable impacts on career outcomes.

IV. Conclusion

My STS research helped me identify key institutional structures that negatively affect first-generation students. This has helped me to focus on designing features in my technical project that will work to increase equity not just for first-generation students but also for other at-risk students. After applying Bronfenbrenner's bioecological systems theory, we can see that by creating an app that is highly usable in our institution, we can support creating an opportunity ecosystem for first-generation students.

I expect to find that my technical project increases clarity in the club discovery process and increases student engagement in CIOs across grounds. In my STS project, I expect to find which institutional practices would most strongly support first-generation students in finding belonging in and increasing access to CIOs. I hope that the combination of my research and technical work will reduce participation gaps in professional organizations between first-generation and continuing-generation students. Lastly, I hope to demonstrate a scalable model for how higher education institutions can create modern engagement systems for their student organizations, not just for first-generation students, but for all students.

V. Bibliography

Abeyta, A. A., & Du, X. (2025). *What makes college meaningful for first-generation students? A quantitative content analysis of first-generation and continuing-generation student writings on why college is purposeful and important. Journal of First-Generation Student Success*, 1–19. <https://doi-org.proxy1.library.virginia.edu/10.1080/26906015.2025.2546412>

Demetriou, C., Meece, J., Eaker-Rich, D., & Powell, C. (2017). The activities, roles, and relationships of successful first-generation college students. *Journal of College Student Development*, 58(1), 19–36. <https://doi.org/10.1353/csd.2017.0001>

Inkelas, K. K., Daver, Z. E., Vogt, K. E., & Leonard, J. B. (2007). Living–learning programs and first-generation college students’ academic and social transition to college. *Research in Higher Education*, 48(4), 403–434. <https://doi.org/10.1007/s11162-006-9031-6>

Ives, J., & Castillo-Montoya, M. (2020). First-generation college students as academic learners: A systematic review. *Review of Educational Research*, 90(2), 139–178. <https://doi.org/10.3102/0034654319899707>

Jackson, D., Lambert, C., Sibson, R., Bridgstock, R., & Tofa, M. (2024). Student employability-building activities: Participation and contribution to graduate outcomes. *Higher Education Research & Development*, 43(6), 1308–1324. <https://doi.org/10.1080/07294360.2024.2325154>

Patel, D., Grubbs, T., Kamath, A., & Shipe, M. (2025, September 24). *Project proposal & plan* [Unpublished class project]. CS 4971: Capstone Practicum II, University of Virginia.

Simpfenderfer, A. D., Wolniak, G. C., Wise, K., et al. (2025). Exploring the associations among major fields of study, pre-professional activities, and post-college job quality in alumnx of liberal arts colleges. *Higher Education*, 90, 667–690. <https://doi.org/10.1007/s10734-024-01342-7>

Soria, K. M., & Cole, A. (2023). Leadership education and first-generation students' social capital development. *Journal of First-Generation Student Success*, 3(1), 1–19. <https://doi-org.proxy1.library.virginia.edu/10.1080/26906015.2023.2171823>

Sudbrock, C., Uribe-Zarain, X., Glaessgen, T. A., Sottile, J., & Finch, K. K. (2024). Examining social capital and first-generation student status: Evidence from a Midwest university. *Journal of Research in Technical Careers*, 8(2). <https://doi.org/10.9741/2578-2118.1155>

Tucker-Smith, O. (2025, November 15). A campus backlash is brewing over Wall Street's Ivy League recruitment drive. *The Wall Street Journal*. <https://www.wsj.com/us-news/education/a-campus-backlash-is-brewing-over-wall-streets-ivy-league-recruitment-drive-c7e6cda9>